Chapter 6

design rules

design rules

Designing for maximum usability
– the goal of interaction design

- Principles of usability
  – general understanding
- Standards and guidelines
  – direction for design
- Design patterns
  – capture and reuse design knowledge

Principles to support usability

Learnability

the ease with which new users can begin effective interaction and achieve maximal performance

Flexibility

the multiplicity of ways the user and system exchange information(eg. Input using Text, audio, gesture

Robustness

the level of support provided to the user in determining successful achievement and assessment of goal-directed behaviour

How well a system supports users in achieving their goals despite mistakes, unexpected inputs

Principles of learnability

Predictability

- determining effect of future actions based on past interaction history
- operation visibility

Example: E-commerce Filter

- Selecting a ”Sort by Price: Low to High” always sorts items accordingly
- Next, previous buttons
- Users know what to expect and can rely on the system

Principles of learnability

Synthesizability

- degree to which users can interpret the results of their actions and integrate that knowledge to improve future interaction
  - Reduces trial-and-error frustration
  - Supports building a mental model of system behavior

- Eg. **User formats text (bold, italic**)
  - Immediate visual feedback shows what changed
  - User can link the action to the result → learns how formatting works

  - **In e-commerce:** User applies a price filter → list updates immediately

    User sees how the filter affects results → understands filter behavior

Principles of learnability (ctd)

Familiarity
- how prior knowledge applies to new system
- guessability; affordance
- Eg. Search, menus…

Generalizability
- extending specific interaction knowledge to new situations
- -eg. fb,twitter,tiktok,etc

Consistency
- likeness in input/output behaviour arising from similar situations or task objectives

Principles of flexibility

Dialogue initiative

- freedom from system imposed constraints on input dialogue eg. Google speech
- system vs. user pre-emptiveness

Multithreading

- ability of system to support user interaction for more than one task at a time
- concurrent vs. interleaving; multimodality
- Concurrent- simultaneously
- Interleaving- one task after the other
- Multimodality- text, audio

Principles of flexibility (ctd)

Substitutivity

- allowing equivalent values of input and output to be substituted for each other
- representation multiplicity; equal opportunity
  - **Eg.** Typing a query, Using voice , Using image search(google lens) for searching
  - Using Password, fingerprint, OTP, social media for login

Customizability

- modifiability of the user interface by user (adaptability) or system (adaptivity)

Principles of robustness

Observability

- ability of user to evaluate the internal state of the system from its perceivable representation
- browsability; defaults; reachability; persistence; operation visibility

•Eg. 80% Loading progress, upload progress, spinning

Recoverability

- ability of user to take corrective action once an error has been recognized
- reachability; forward/backward recovery; commensurate effort

Principles of robustness (ctd)

Responsiveness

- how the user perceives the rate of communication with the system
- Stability

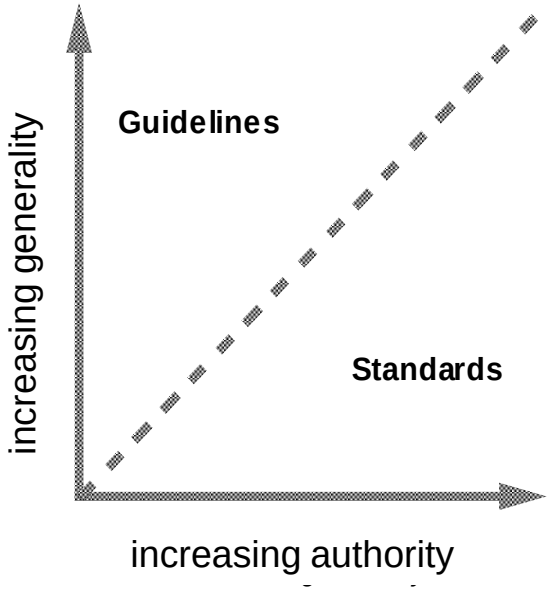
Task conformance

- degree to which system services support all of the user's tasks
- task completeness; task adequacy

Using design rules

Design rules

- suggest how to increase usability
- differ in generality and authority



Standards

- set by national or international bodies to ensure compliance by a large community of designers
- standards require sound underlying theory and slowly changing technology
- hardware standards more common than software high authority and low level of detail
- ISO 9241 defines usability as effectiveness, efficiency and satisfaction with which users accomplish tasks

Standards

Reading assignment

- Explore the following primary global body that creates and publishes official HCI standards:
  1. **ISO 9241** (Ergonomics of Human–Computer Interaction)
  2. **ISO 25010** (Software Quality including usability)
  3. **ISO 13407** (Human-Centered Design Process)

Guidelines

- More suggestive and general
- many textbooks and reports full of guidelines
- abstract guidelines (principles) applicable during early life cycle activities
- detailed guidelines (style guides) applicable during later life cycle activities
- understanding justification for guidelines aids in resolving conflicts

Golden rules and heuristics

- "Broad brush" design rules
- Useful check list for good design
- Better design using these than using nothing!
- Different collections e.g.
  - Nielsen's 10 Heuristics
  - Shneiderman's 8 Golden Rules
  - Norman's 7 Principles

Shneiderman's 8 Golden Rules

1. *Strive for consistency*
2. *Enable frequent users to use shortcuts*
3. *Offer informative feedback*
4. *Design dialogs to yield closure*
5. *Offer error prevention and simple error handling*
6. *Permit easy reversal of actions*
7. *Support internal locus of control(users decide what to do ,where to go, not the system)*
8. *Reduce short-term memory load(recall and recognition)*

Norman's 7 Principles

1. *Use both knowledge in the world(best practice) and knowledge in the head.*
2. *Simplify the structure of tasks.*
3. *Make things visible: bridge the gulfs of Execution and Evaluation.*
   - ***gulfs of Execution*** *:*the gap between what the user **wants to do** and what the system **allows the user to do**
   - ***gulfs of Execution:*** the gap between what the system **does** and how well the user can **understand and confirm what happened**.
4. *Get the mappings right*
   - designing controls and interface elements so that the relationship between a user's action and the system's response is **natural, intuitive, and easy to predict(**e.g submit, next, back buttons**)**
   - Poor mapping—*such as a control that behaves opposite to what users expect—causes confusion, errors, and cognitive load. Good mapping reduces mental effort, speeds learning, and makes interaction feel effortless and transparent*.

Norman's 7 Principles

5. Exploit the power of constraints, both natural and artificial.

- Limiting user actions to prevent errors.
- Improves safety and reduces mistakes

Examples:

- Disabled buttons until required fields are filled
- Date pickers preventing invalid dates
- Input validation restricting wrong formats

6. Design for error.

7. When all else fails, standardize.

- if no single design option is clearly best, choose one and apply it consistently across the system to improve usability and reduce user confusion.
- **Eg Labels should be above or beside input fields**

Name:_____ **Name:**

_____

HCI design patterns

- An approach to **reusing knowledge** about successful design solutions
- Originated in architecture: Alexander
- A pattern is an invariant solution to a **recurrent problem** within a specific context.
- Examples
  - Light on Two Sides of Every Room (architecture)
- Patterns do not exist in isolation but are linked to other patterns in *languages* which enable complete designs to be generated

HCI design patterns (cont.)

- Characteristics of patterns
  - capture design practice not theory
  - capture the essential common properties of good examples of design
  - represent design knowledge at varying levels: *social, organisational, conceptual, detailed*
  - embody values and can express what is humane in interface design
  - are intuitive and readable and can therefore be used for communication between all stakeholders
  - a pattern language should be generative and assist in the development of complete designs.

HCI design patterns (cont.)

- Some Examples of HCI design patterns

1. System Status Visibility

Examples:

- Loading spinners while processing
- Progress bars during file uploads
- "Saved successfully" messages after form submission

2. Affordance: Interface elements visually suggest how they should be used

Examples:

- Buttons that look clickable
- Sliders that invite dragging
- Text boxes with visible input areas

3. Recognition Rather Than Recall

Examples:

- Dropdown menus and toolbars
- Autocomplete suggestions
- Visible icons instead of typed commands

Summary

Principles for usability

- repeatable design for usability relies on **maximizing benefit of one good design** by abstracting out the general properties which can direct purposeful design
- The success of designing for usability requires **both creative insight (new paradigms) and purposeful principled practice**

Using design rules

- standards and guidelines to direct design activity